

Control Flow Analysis - With & Without ACP
22-Agent Pipeline | 35 Governance Controls | End-to-End Flow

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Executive Summary

The Personal Loan Underwriting System uses **22 AI agents** across **5 phases** to process loan applications end-to-end: intake, extraction, risk assessment, credit decision, and disbursement.

Without governance, these agents operate as a **black box** - no audit trail, no cost control, no compliance mapping, no bias detection, no kill switch.

ACP Governance Platform wraps the entire pipeline with **35 controls** in a single line of code:

```
workflow = govern(workflow, "loan-underwriting-22", bu="lending", budget=5.0)
```

This document compares the system **WITH** and **WITHOUT** ACP governance, mapping every control to specific agents and regulatory frameworks (RBI FREE-AI, GDPR, EU AI Act, NIST, OWASP, UK AI).

WITHOUT ACP Governance - Risk Exposure

When AI agents run without governance controls, the following risks are unmitigated:

Risk Area	What Happens	Business Impact
No Audit Trail	Agent decisions are not logged. No way to explain why a loan was denied.	RBI non-compliance. Customer complaints. Legal exposure.
No Cost Control	Agents make unlimited LLM API calls. No budget enforcement.	Runaway costs. \$50K+ monthly bills with 22 agents.
No PII Protection	Aadhaar, PAN, income data flows through all agents unmasked.	Severe data breach risk. GDPR/DPDPA violation. Fines up to 4% revenue.
No Bias Detection	Credit decisions may discriminate by region, income band, Regulator.	Regulatory action. Reputational damage. Portfolio skew.
No Human Oversight	TIER_3 agents auto-execute disbursements with no approval.	Unauthorized fund transfers. Fraud risk.
No Compliance Mapping	No mapping to RBI FREE-AI, GDPR, EU AI Act, NIST.	Audit failure. Cannot prove compliance to regulators.
No Rate Limiting	Unlimited calls/sec to CIBIL, Experian APIs.	API bans. Service degradation. SLA violations.
No Explainability	Cannot explain why a decision was made. Black box.	RBI mandate violation. Right to explanation denied.
No Injection Defense	Prompt injection can manipulate agent decisions.	Fraudulent approvals. Credit loss.
No Kill Switch	Rogue agent cannot be stopped.	Cascading failures. Cannot contain incidents.

WITH ACP Governance - 35 Controls Active

Key controls and their loan underwriting application:

#	Control	What It Does	Loan Example
Q1	Loop Guard	Max iterations + dead-man switch	Stops after 50 iterations or 20-step trigger
Q2	Policy Engine	Business rule evaluation	DTI > 0.65 = auto-deny. Age 21-58.
Q4	Tracing	Full audit trail via LangFuse	Every decision logged with timestamp
Q6	Autonomy Tier	Human approval gates	TIER_2: CreditDecision needs review
Q8	Security	Credential injection detection	Blocks API keys in agent flow
Q11	Cost Control	3-tier budget enforcement	\$5.00/app. Warn 80%, block 200%.
Q13	Compliance	Framework mapping	RBI, GDPR, EU AI Act, NIST, OWASP
Q14	Fairness	Bias detection	Monitors by region, income, gender
Q15	Explainability	Chain-of-thought logging	XAI generates human-readable reasons
Q18	Tenant Isolation	Multi-BU separation	Only lending BU accesses credit data
Q19	Consent	GDPR consent tracking	Verifies consent before Bureau Pull
Q22	SLA	Latency monitoring	Bureau calls must complete in <3s
Q25	PII Masking	Auto-mask sensitive data	Aadhaar XXXX-XXXX-1234 in logs
Q28	Hallucination Guard	Grounding check	Decision must be data-supported
Q29	Injection Firewall	Prompt injection defense	Blocks adversarial manipulation
Q34	Confidentiality	Auto-classification	Marks loan data as RESTRICTED
Q35	Data Flow	Cross-agent taint tracking	Tracks PII from intake to disburse

Loan Application Flow - 5 Phases, 22 Agents

Phase 1: Intake

Agents: LeadQualification --> DocumentCollection

Controls: Q1, Q2, Q4, Q19, Q25, Q29, Q33, Q34

Phase 2: Extraction (parallel)

Agents: DocClassifier --> OCR --> DataNormalizer | BureauPull | EmployerVerify | SalariedIncome | SelfEmployedIncome | AccountAggregator

Controls: Q8, Q9, Q11, Q14, Q16, Q18, Q22, Q30, Q32, Q35

Phase 3: Risk Assessment (parallel)

Agents: RiskModeler | IncomeStability | DebtBurden | IdentityFraud | DocTampering --> ScoreAggregator

Controls: Q2, Q7, Q10, Q14, Q15, Q17, Q21, Q23

Phase 4: Decision

Agents: ComplianceCheck | Pricing --> CreditDecision --> OfferGenerator | XAIExplainer

Controls: Q5, Q6, Q12, Q13, Q15, Q28, Q31, Q32, Q34

Phase 5: Fulfillment

Agents: Disbursement

Controls: Q5, Q6, Q11, Q20, Q24

Controls Fired per Agent - Complete Matrix

Agent	Phase	Tier	Key Controls
LeadQualification	Intake	TIER_1	Q1, Q2, Q4, Q29, Q33
DocumentCollection	Intake	TIER_1	Q19, Q25, Q34
DocClassifier	Extraction	TIER_1	Q4, Q30
OCRExtractor	Extraction	TIER_1	Q8, Q32
DataNormalizer	Extraction	TIER_1	Q9, Q35
BureauPull	Extraction	TIER_1	Q11, Q16, Q18, Q22
EmployerVerify	Extraction	TIER_1	Q4, Q16
SalariedIncome	Extraction	TIER_1	Q14, Q15
SelfEmployedIncome	Extraction	TIER_1	Q14, Q15
AccountAggregator	Extraction	TIER_1	Q19, Q35
RiskModeler	Risk	TIER_2	Q10, Q14, Q15, Q17
IncomeStability	Risk	TIER_1	Q7, Q10
DebtBurden	Risk	TIER_1	Q2, Q9
IdentityFraud	Risk	TIER_3	Q8, Q23, Q29
DocTampering	Risk	TIER_3	Q23, Q34
ScoreAggregator	Risk	TIER_2	Q1, Q11, Q21
ComplianceCheck	Decision	TIER_2	Q4, Q13, Q32
Pricing	Decision	TIER_2	Q2, Q7, Q11
CreditDecision	Decision	TIER_2	Q5, Q6, Q12, Q28
OfferGenerator	Decision	TIER_1	Q31, Q34
XAIExplainer	Decision	TIER_1	Q15, Q32
Disbursement	Fulfillment	TIER_3	Q5, Q6, Q11, Q20, Q24

8 Controls Active on Every Agent Call

These controls fire on every single tool call across all 22 agents:

Control	What It Does
Q1 Loop Guard	Increment iteration count, enforce max 50, dead-man switch at step 20
Q4 Tracing	Log every call with timestamp, agent name, inputs, outputs, cost
Q7 Quality	Track latency, accuracy, override rate for each agent
Q11 Cost	Accumulate cost per call, check against \$5.00 budget (warn 80%, block 200%)
Q22 SLA	Track p95 latency vs 3000ms target, error budget monitoring
Q25 PII Scan	Scan all inputs/outputs for PAN, Aadhaar, credit card patterns
Q33 Rate Limiter	Token bucket enforcement, max 60 calls/minute per agent
Q34 Confidentiality	Auto-classify every output as PUBLIC/INTERNAL/CONFIDENTIAL/RESTRICTED

What the CIO Sees - ACP Governance Dashboard

After a loan application is processed, the CIO can see:

1. **Governance Map** - Visual graph of 22 agents with real-time status, budget usage, control health
2. **Agent Actions** - Pause, Stop, Kill, Resume, De-escalate buttons on each agent
3. **Controls Page** - All 35 controls with active/warning/inactive status
4. **Audit Log** - Every tool call logged: allowed/blocked, reason, cost, tokens
5. **Compliance Scores** - RBI FREE-AI, GDPR, EU AI Act, NIST AI RMF, UK AI, OWASP LLM
6. **Cost Center** - Total cost per application, per agent, per day
7. **HITL Queue** - Escalated applications waiting for human review
8. **Approvals** - Pending approval requests for supervised (TIER_2) decisions
9. **Activity Log** - All CIO actions (pause, kill, de-escalate) logged with timestamp

Dashboard URL: <https://dashboard.acpgovernance.com/governance-map>

Regulatory Compliance - 6 Frameworks Mapped

Framework	Key Requirements	ACP Controls
RBI FREE-AI	Fairness, Reliability, Explainability, Ethics, Accountability	Q14, Q7, Q15, Q13, Q4
GDPR	Right to explanation, Data minimization, Consent, Right to erasure	Q15, Q19, Q31, Q25
EU AI Act	Risk classification, Human oversight, Transparency, Robustness	Q6, Q15, Q23, Q32
NIST AI RMF	Govern, Map, Measure, Manage	Q2, Q13, Q7, Q1
OWASP LLM Top 10	Prompt injection, Data leakage, Insecure output	Q29, Q25, Q8, Q9
UK AI Framework	Safety, Transparency, Fairness, Accountability, Contestability	Q8, Q15, Q14, Q4, Q12

Summary - With vs Without ACP

Without ACP	With ACP
Black box decisions	Full audit trail + explainability (Q4, Q15)
Unlimited costs	\$5.00 budget with 3-tier enforcement (Q11)
PII exposure	Auto-masking + GDPR consent management (Q19, Q25, Q31)
No bias detection	Real-time fairness monitoring (Q14)
No human oversight	TIER-based approval gates + de-escalation (Q6)
Zero compliance proof	6 framework mapping with scores (Q13)
Vulnerable to injection	Prompt injection firewall + red team testing (Q23, Q29)
No kill switch	Pause / Stop / Kill / Resume from Governance Map
No cost visibility	Per-agent, per-application cost tracking
Uncontrolled agents	35 governance controls, all active, all audited

ACP Governance Platform - One line of code. 35 controls. Complete AI governance.

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